

Successful mepolizumab treatment for DRESS-induced refractory eosinophilic myocarditis and concurrent thyroiditis.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) is a potentially life-threatening adverse drug reaction with a mortality rate of 10%. Interstitial nephritis, pneumonitis, myocarditis, meningitis, thyroiditis and pancreatitis are major causes of morbidity and mortality in this syndrome. Cessation of offending medication is paramount. There is paucity in high quality prospective studies guiding the treatment of DRESS, and there are no published therapeutic clinical trials in the treatment of corticosteroid refractory hypersensitivity myocarditis. The authors present a unique case of ciprofloxacin-induced DRESS with concurrent thyroiditis and refractory eosinophilic myocarditis that required mepolizumab and multiple immunosuppressants for successful treatment.

Drug Rash with Eosinophilia and Systemic Symptoms (DRESS) Syndrome Induced by Imatinib.

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Imatinib is the treatment of choice in patients with locally advanced or metastatic gastrointestinal stromal tumours (GIST). Clinical tolerance of imatinib is excellent except for the common adverse drug reaction (ADR). Drug rash with eosinophilia and systemic symptoms (DRESS syndrome) is a severe, potentially life-threatening drug-induced hypersensitivity reaction, characterized by cutaneous eruptions, fever, diffuse lymphadenopathy, along with eosinophilia, and elevated liver function tests. This ADR is rarely reported with imatinib. Only four cases of DRESS syndrome associated with imatinib have already been published. We report an exceptional case of DRESS syndrome associated with imatinib in a 46 year-old woman with GIST. Two weeks after she had started imatinib therapy, she developed a skin rash, with eosinophilia and elevated liver tests. Plasma level of imatinib was within the therapeutic range. Imatinib was immediately discontinued. A favourable outcome was slowly observed after the drug had been stopped. This case was scored three according to the European Registry of Severe Cutaneous Adverse Reactions Study Group (RegiSCAR). The Naranjo score for imatinib was five (probable).

Pulmonary Manifestations of Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) Syndrome: A Systematic Review.

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The syndrome of drug reaction with eosinophilia and systemic symptoms (DRESS) is a rare, yet potentially fatal hypersensitivity reaction, most commonly associated with anticonvulsants, sulfonamides, and allopurinol. The reaction commonly manifests as a febrile skin eruption with lymphadenopathy and malaise between two and eight weeks following drug exposure. Internal organ involvement occurs in close to 90 percent of patients, and multiple organs may be involved in approximately half of those affected (most commonly the liver, kidney, and lung). Its long latency period and its variable clinical pattern of presentation have earned it the moniker of "the great mimicker," with delays in diagnosis leading to higher morbidity and mortality. Although less commonly affected in DRESS syndrome, lung involvement is associated with more severe clinical course and potentially worse outcome. Pulmonary

symptoms may precede development of the other more common symptoms and signs of the syndrome, or they might develop later in the course of the disease. Lung involvement in DRESS presents with a plethora of manifestations from mild cough or dyspnea with nonspecific interstitial changes on chest imaging to acute respiratory distress syndrome (ARDS) with life-threatening hypoxic respiratory failure. We performed a systematic review of literature from the PubMed database and selected cases of definite DRESS syndrome as defined by the European Registry of Severe Cutaneous Adverse Reactions (RegiSCAR) with a score of 6 or more who also had pulmonary involvement. Demographic data, pattern of lung involvement, culprit medication, latency period, laboratory findings, therapy, and outcome were described and compared with the literature. The most common pulmonary radiographic findings in DRESS were interstitial infiltrates in 50% of cases, followed by acute respiratory distress syndrome (ARDS) 31%. Symptoms of cough and shortness of breath (SOB) were present in 72% of patients at the time of presentation. SOB was the more common presenting symptom (81%) compared to cough (19%). In 95% of cases, another visceral organ was involved (most commonly liver or kidneys). 45% of cases were initially misdiagnosed as pneumonia and were treated with empiric antimicrobials. In a multivariate regression, a latency of 30 days or less and an age of 60 or less were associated with development of ARDS. Gender and eosinophil count were not associated with severity of pulmonary manifestations. All patients recovered, and in the vast majority of cases (95%), parenteral steroids were used for treatment in addition to supportive care and symptomatic management. Albeit rare, DRESS is a potentially life-threatening syndrome which may present with a myriad of pulmonary signs and symptoms. Pulmonary manifestations are less common but are typically seen in more severe cases. Pulmonary manifestations may be a presenting sign of DRESS, and timely recognition is important in order to stop offending medication and decrease morbidity and mortality.

Demystifying drug reaction with eosinophilia and systemic symptoms (DRESS): a review of the literature and guidelines for management.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) is a severe, adverse drug reaction that is notoriously complex in both its presentation and treatment. Although early diagnosis and cessation of the causative agent are universally accepted as the initial interventions for DRESS, the subsequent management lacks a standardized approach. Historically, systemic steroids have been used as first-line treatment, but there is debate about the optimal dosing and route of administration, and evidence persists on the long-term complications associated with steroid use. Novel treatment approaches with targeted therapy, cyclosporine, intravenous immunoglobulin, and plasmapheresis have been gaining interest as alternative mono- and adjuvant therapies, but their use has yet to be supported by clinical trials. This narrative review provides a summary of the current knowledge of DRESS, with a focus on clinical management. The various mono- and adjuvant therapy options are discussed, with literature-supported suggestions for their optimal use in clinical practice. The risks for relapses, viral reactivation, and long-term complications are also considered. The PubMed and Medline databases were searched for articles on DRESS, published between January 1, 2008, and May 1, 2023. 334 articles met the inclusion criteria. Based on the literature, a DRESS management tool with step-by-step guidance is provided. Further suggestions for management are woven throughout this review, giving clinicians a toolbelt of resources with which to approach diagnosis, treatment, and follow-up.

DRESS syndrome: an interaction between drugs, latent viruses, and the immune system.

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Drug-induced hypersensitivity syndrome, also known as DRESS syndrome, is a serious and potentially fatal reaction that occurs in response to prolonged use (generally between 14 and 60 days) of certain drugs, and which has no predilection for gender or age group. It is believed that DRESS syndrome has a genetic basis and results from the interaction between metabolites of certain pharmacological groups, reactivation of latent viruses (especially from the Herpesviridae family), and a cellular immune response. The classic manifestation of DRESS syndrome includes a generalized rash accompanied by fever, eosinophilia, lymphadenopathy, and systemic involvement such as hepatitis, nephritis, or pneumonitis. With the continuous increase in the availability of drugs and the aging of the population, there is a favorable scenario for the development of adverse drug reactions. Physicians should be prepared for the early diagnosis of DRESS syndrome, the identification and immediate suspension of the drug involved, and also manage systemic involvement, which may require prolonged immunosuppressive therapy. This article provides an update on the clinical, physiopathological and therapeutic aspects of DRESS syndrome.

Drug reaction with eosinophilia and systemic symptoms (DRESS): a clinical update and review of current thinking.

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Summary Drug reaction with eosinophilia and systemic symptoms (DRESS) describes a severe medication-induced adverse reaction, which has cutaneous, haematological and solid-organ features. It is one of the triad of life-threatening drug hypersensitivity dermatoses, along with acute generalized exanthematous pustulosis (AGEP) and Stevens-Johnson syndrome/toxic epidermal necrolysis (SJS/TEN). In this article, we discuss several controversies that surround DRESS, including problems with nomenclature and the lack of consensus in diagnostic criteria.

[Drug reaction with eosinophilia and systemic symptoms (DRESS): a hypersensitivity reaction with various symptoms].

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Drug reaction with eosinophilia and systemic symptoms (DRESS) is an illness which is difficult to diagnose because of its various symptoms. In our case, a patient with small spotted exanthema with nearly erythroderma and eosinophilia presented to the emergency room. Systemic steroid therapy was started on suspicion of a drug reaction. Over the course of time, the patient's general condition deteriorated significantly and the patient developed cholecystitis, *Staphylococcus aureus* bacteremia, pneumonitis and cytomegalovirus reactivation. With this case report, we want to show that DRESS is a disease that is difficult to treat and can develop after a long delay.

Drug Reaction with Eosinophilia and Systemic Symptoms: An Update and Review of Recent Literature.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome, is also known as drug induced hypersensitivity syndrome and by various other names. It is now recognised as one of the severe cutaneous adverse reaction (SCAR) and can be potentially life-threatening. Historically, it was most frequently linked with phenytoin and was initially described as phenytoin hypersensitivity syndrome. However, it was later found to be caused by various other medications. Anticonvulsants and sulfonamides are the most common offender. Characteristically DRESS has a latent period of 2 to 6 weeks. The pathophysiology remains incompletely understood but involves reactivation of viruses and activation of lymphocyte. It is manifested most commonly with a morbilliform cutaneous eruption with fever and lymphadenopathy. The severity of this syndrome is related to the systemic involvement, which can result in multi-organ failure. Most important step in the management of DRESS is early diagnosis and immediate cessation of the suspected offending drug. Patients of DRESS syndrome should be managed in an intensive care set up for appropriate supportive care and infection control. Topical corticosteroids can give symptomatic relief, but systemic therapy with steroid and other immunosuppressant is usually required.

Successful Treatment of Antiepileptic Drug-Induced DRESS Syndrome with Pulse Methylprednisolone.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is a rare but potentially life-threatening syndrome characterized by skin rash, fever, lymph node enlargement, and involvement of internal organs. It is most commonly induced by aromatic anticonvulsants and antibiotics. Nonaromatic anticonvulsants are rarely encountered as the causes of DRESS syndrome. In the present report, three discrete cases with DRESS syndrome developing due to three antiepileptic drugs, including valproic acid (nonaromatic), carbamazepine (aromatic), and lamotrigine (aromatic), and their treatment modalities were aimed to be discussed in light of the literature. To the best of our knowledge, our cases are the first children to be treated with pulse methylprednisolone in the literature.

Clozapine-related drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome: a systematic review.

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The Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) syndrome is a severe, multiorgan, and potentially life-threatening drug-induced hypersensitivity reaction, linked to several common drugs, including antiepileptics, antibiotics, and several psychotropic drugs, including clozapine. Due to the importance of clozapine in the management of treatment-resistant schizophrenia, a systematic review and characterization of clozapine-related DRESS syndrome is long overdue. This systematic review was conducted following PRISMA guidelines. PubMed, Embase, PsychINFO, and the Cochrane Library databases were independently reviewed up to 1 November 2019 for articles reporting

clozapine-related DRESS syndrome cases. The RegiSCAR score system was applied to systematically characterize the clinical presentations of selected studies. Clozapine-related DRESS syndrome was reported in six patients from four articles. Five patients received polypharmacy. Skin rash and liver involvement with elevated liver enzymes were very common. No fatal cases were found. Treatment mainly included clozapine discontinuation and immunosuppression. The mismatch between incidences of DRESS with other responsible drugs, the common misdiagnosis of this syndrome, and the fact that an extensive literature search only identified six cases suggests that clozapine-related DRESS may be overlooked. It is, therefore, necessary to optimize diagnostic strategies to identify immune-related side effects of clozapine.
